

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 1(a) | <p><i>any <b>five</b> from:</i></p> <ol style="list-style-type: none"> <li>1 (ACTH) binds to receptor on cell surface membrane (of adrenal gland cells) ;</li> <li>2 G protein activated ;</li> <li>3 adenylyl cyclase activated ;</li> <li>4 cyclic AMP made from ATP ;</li> <li>5 (cyclic AMP) second messenger ;</li> <li>6 (cyclic AMP) binds to kinase (enzyme) ;</li> <li>7 <i>ref. to</i> enzyme cascade / activation of enzymes by phosphorylation ;</li> <li>8 amplifies signal ;</li> <li>9 (amplification of signal) results in synthesis of cortisol ;</li> </ol> | <b>5</b> |
| 1(b) | <u>negative feedback</u> ;                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>1</b> |
| 1(c) | <p><i>any <b>two</b> from:</i></p> <ol style="list-style-type: none"> <li>1 uncontrolled mitosis / continuous cell cycle / cell cycle checkpoints not controlled ;</li> <li>2 abnormal mass of cells formed ;</li> <li>3 no programmed cell death / no apoptosis / cells immortal ;</li> <li>4 AVP ; e.g. mutation of, tumour suppressor gene / (proto)oncogene</li> </ol>                                                                                                                                                                                                    | <b>2</b> |

| Question | Answer                                                                                                                                                                                                                                                                                  | Marks |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 1(d)     | <p><i>any <b>one</b> from:</i></p> <ol style="list-style-type: none"> <li>1 decrease in water potential of the blood ;</li> <li>2 eating food with high salt content ;</li> <li>3 excessive panting (because of increased body temperature) ;</li> <li>4 AVP ; e.g. diabetes</li> </ol> | 1     |

| Question | Answer | Marks |
|----------|--------|-------|
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